

Technical Systems & Architecture Manual

ENGINEERING OPERATIONS • VECTOR DB & KNOWLEDGE OS

1. Vector Database Infrastructure

The system leverages FAISS vector indices for fast approximate nearest neighbor (ANN) search over dense document embeddings. Embeddings are synchronized via distributed pipelines.

2. Ingestion & Preprocessing Pipeline

Documents undergo chunking, semantic parsing, metadata tagging, and dense embedding generation prior to vector indexing. Incremental indexing runs on scheduled micro-batches.

3. API Integration & Retrieval

Retrieval queries execute a hybrid search combining dense vector embeddings with sparse BM25 term matching, re-ranked via cross-encoder models.